Write-Ahead Logging and Recovery Algorithms for ACID Guarantees

Write-Ahead Logging (WAL) is the cornerstone mechanism by which modern relational databases (including SQLite and PostgreSQL) guarantee durability and atomicity under unexpected system crashes or power failures. The fundamental WAL rule dictates that any modification to a database data page must be appended to the append-only log on non-volatile disk before the corresponding page can be flushed from the in-memory buffer pool to the main database file.

During crash recovery, the database executes the classic ARIES recovery protocol, consisting of three sequential phases: 1. Analysis phase: Scans the log forward from the last checkpoint to identify dirty pages in buffer pool and active uncommitted transactions at the moment of the crash. 2. Redo phase: Replays history forward from the earliest unwritten log record, repeating every logged operation to bring the database state back to the exact instant of the crash. 3. Undo phase: Scans backward, undoing the operations of all transactions that were still active when the crash occurred.